Control Barrier Functions

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Introduction

Robotic control functions often encounter situations where some options would have significant negative consequences, such as a self-driving car crashing, or a machining robot crushing its payload. In OSHA's guidelines for robotic system safety [1], employers and workers who design robotic systems are expected to eliminate or minimize robotic hazards. While it is optimal to introduce physical limits to the system that prevent unsafe conditions (see Figure 1), this is not always possible. Instead, mathematical structures within the control scheme can provide "safety-related parts of the control system" to enhance limitations.

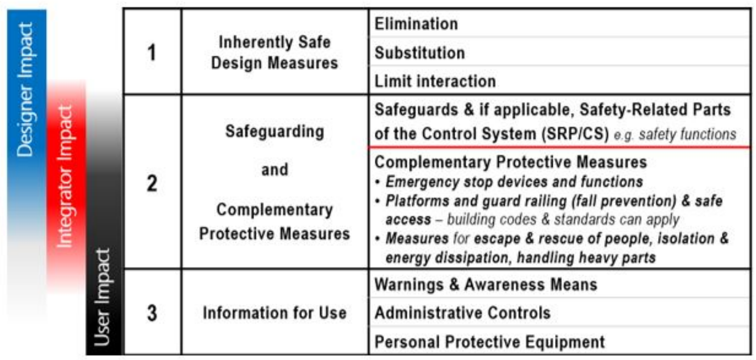

Figure 1: OSHA's Hierarchy of Controls 3-step method for applying safe robotics

Control Barrier Functions (CBF) [2] are a methodology for creating "safe zones" or "danger zones" in control systems. Our proposal is to explore applications of CBFs on a variety of reinforcement learning systems. We will be exploring how CBFs can be used to enhance safety in the following systems:

1. A five-axis machining arm performing a pick-and-place activity while avoiding obstacles.
2. A drone flying in a Gymnasium RL with restricted flight areas.
3. A satellite conducting time-optimal slewing maneuvers while strictly enforcing a keep-out zone.

Goals and Objectives

Primary Goal

Each of the systems listed in the introduction will implement CBFs to avoid a negative outcome. Each method will compare at least one CBF influenced controller with at least one non-CBF controller and evaluate the relative training time, likelihood or severity of the unsafe condition, and the efficiency of the learned control scheme.

1. Five-axis Machining Arm: The tool will be implemented using the gymnasium-robotics environment. The goal is to ensure that the machining arm can move objects between two locations while avoiding specific obstacles [6].
2. Safety-Filtered Drone Flight: Implementation of a CBF filter for a drone using the gym-pybullet-drones [10] environment. The goal is to ensure the drone maintains "Well Clear" distances from obstacles or boundaries [5].

3. Satellite Attitude Control: Apply Reinforcement Learning coupled with High-Order CBF (HOCBF) to perform time-optimal slewing maneuvers while strictly enforcing a keep-out zone, protecting sensitive onboard instruments from direct solar radiation [9].

Backup Goal

In the situation where one or more members are unable to implement the CBF control for their chosen system, they will have an opportunity to collaborate with one of the other members to expand upon their chosen methodologies. Some additional CBF strategies, such as constraint-guided formulations [7] and CBF-RL [8] can be applied to the same system to compare the strategies' efficacy, training time, and safety.

Stretch Goal

Each member has identified one or more additional algorithms which could expand or augment the CBF functionality.

1. Five-axis Machining Arm: Introduce dynamic, moving obstacles into the five-axis machining arm system to simulate environments in close proximity to other robots.
2. Safety-Filtered Drone Flight: Integrate SINDy to learn a data-driven model of the drone dynamics from simulation data, and use this learned model to construct a more accurate Control Barrier Function (CBF) constraint.
3. Satellite Attitude Control: Explore Safe RL algorithms (like PPO-Lagrangian) or fine-tuning the GP-HOCBF confidence bounds (β) and penalty parameters to be even more conservative against highly dynamic unmodeled disturbances.

Division of Responsibilities

- Sam Zimmerman will lead the five-axis machining arm system, and will maintain the GIT repository and resolve code conflicts.
- Adriano Canolla will lead the Drone PyBullet implementation, and will act as project scheduler & coordinator.
- Tyler Darwin will focus on satellite attitude control with a keep-out zone, and will manage and coordinate the drafting and submission of the project reports.
- All team members will contribute to the data synthesis, final integration, and drafting of the reports.

CRediT statement

Adriano Canolla - Conceptualization, Investigation, Resources for drone flight system, Writing – review & editing, Supervision

Sam Zimmerman - Conceptualization, Investigation, Resources for machining arm system, Writing – original draft, Writing – review & editing

Tyler Darwin - Conceptualization, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing

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